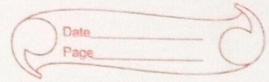


Day - 1

6/4/26



1. Overview of Network Communications (Protocols)

1. Data Communication -

It is exchange of data between two Nodes via some form of link (Transmission medium) such as cable

2. Data Flow -

1] Simplex -

Communication is unidirectional.
one device can transmit & other can will receive

Ex - keyboards, traditional Monitor.

2] Half - Duplex -

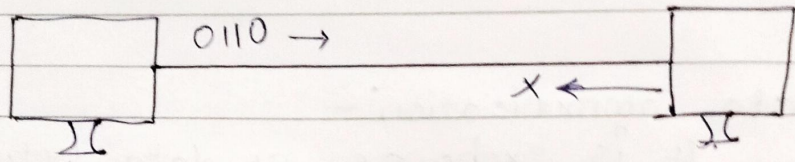
Communication is in both direction but Not at same time.

Ex - Walkie-talkies.

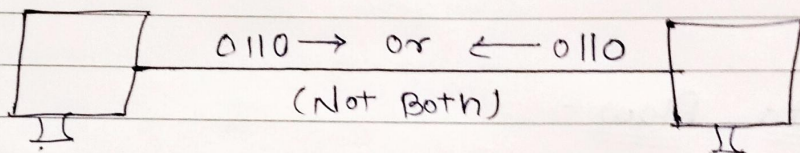
3] Full Duplex / Duplex -

Communication happen in both direction simultaneously.

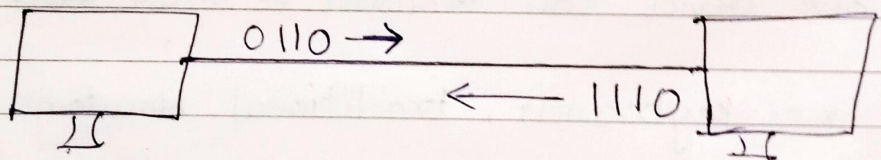
1] Simplex -



2] Half Duplex -



3] Full Duplex -



3*] Protocols

All Communication Schemes will have the following in Common.

1] Source or Sender

2] Destination or receiver.

3] channel or media.

Protocol -

Rules or protocols govern all methods of communication.

Protocol Determines -

- 1] what is communicated
- 2] How it is communicated
- 3] when it is communicated.

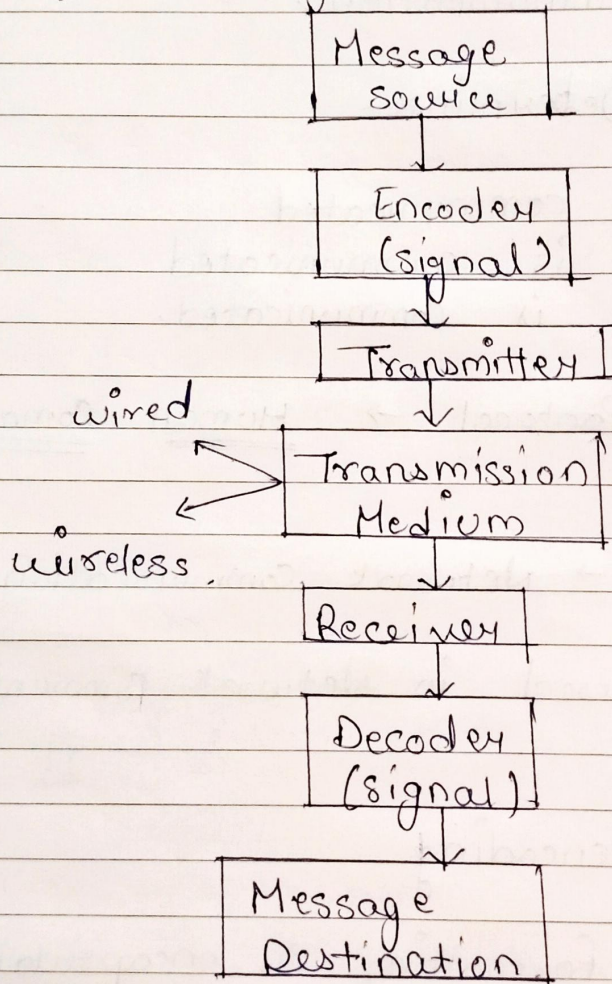
Ex of Protocol → Human Communication

⇒ Protocols → Network Communication.

Protocols used in Network Communication also define.

- 1] Message encoding
- 2] Message formatting & encapsulation
- 3] Message timing
- 4] Message size
- 5] Message Delivery options.

1] Message encoding -



2] Message formatting & Encapsulation

- Agreed format.
- Encapsulated the information to identify source & destination

3] Message Size -

Human Break long Message into smaller parts or sentences.

4] Message Timing -

- Flow Control
- Response Timeout.

5] Message Delivery options

- 1] unicast (one-to-one)
- 2] Multicast (one-to-few (selected))
- 3] Broadcast (one-to-all)

6/4/26.

2. OSI Model

- OSI → open System Interconnection Model.
- It is a guide
- This is not OSI Protocol Suite.
- There are unique Protocol at every Layer.

Layer 7

Application

Layer 6

Presentation

Layer 5

Session

Layer 4

Transport

Layer 3

Network

Layer 2

Data Link

Layer 1

Physical.

1] Layer - 1 - Physical.

- "The Physics of Network"
- It contains cabling, signaling, connectors etc
- This layer isn't about Protocol

2] Layer 2 - Data Link layer.

- "switching" layer.
- The Foundation of Communication at data link.

3] Layer - 3 - Network layer.

- MAC (Media Access Control) protocol, & Address on Ethernet.

4] Network layer. - Layer - 3

- The "routing" layer.
- Internet Protocol (IP)
- fragments frame to traverse different Network.

4] Layer - 4 - Transport

- The "Post office" Layer.

- Protocols → 1] TCP (Transmission Control Protocol)
2] UDP (User Datagram Protocol)

5] Layer 5 - Session.

- Communication Management between devices.

- Control protocol, tunneling protocol

6] Layer 6 - Presentation

- character encoding

- Application Encryption

- often combined with Application layer.

#] Layer 7 - Application

- Layer we see. !
- ~~Protocol~~ HTTP, FTP, POP3, DNS.

Real-world to OSI

Layer 7 Application	Your eyes (HTTP, FTP, POP3)
Layer 6 Presentation	Application Encryption (SSL/TLS)
Layer 5 Session	Control Protocol, Tunneling Protocol
Layer 4 Transport	TCP, UDP
Layer 3 Network	IP Addressing, Router, Packet
Layer 2 Data Link	MAC Address, Frames, Switch
Data Layer 1 Physical	cable, signal, Connectors.

Cyberattacks at every OSI Layer

Layer 7 Application	SQL injection, Cross-site scripting (xss), Malware injection, Phishing.
Layer 6 Presentation	SSL/TLS attacks, Encryption attacks, Data format exploitation
Layer 5 Session	Session hijacking, Session fixation
Layer 4 Transport	TCP SYN flood, UDP flood, Session hijacking, TCP reset attack
Layer 3 Network	IP spoofing, ICMP flooding, Smurf attack, Routing attack
Layer 2 Data link	MAC spoofing, ARP spoofing/Poisonous, VLAN hopping, Switch port stealing
Layer 1 Physical	Cable tapping, Hardware theft, Signal Jamming